

RAGHAVENDRA DINESH

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SUMMARY

Deep Learning Researcher with 3+ years of experience in deep learning, computer vision, and multimodal models, gained through roles at ASU and Bosch. Skilled in developing innovative AI solutions, with a strong focus on Generative AI, LLMs, and model optimization for high-impact applications.

PROFESSIONAL EXPERIENCE

Deep Learning Researcher

June 2024 - Present

Arizona State University

Plant Image based Health Diagnostics System with BLIP-2 Image Captioning using Vicuna 7B and Gemini API

- Achieved **98% classification accuracy** on PlantVillage Dataset with **ResNet50 classifier** quantized using **Onnx** and **TFLite**.
- Obtained **Rouge-N score of 0.63** on text generated from **Gemini API** passing prompts based on image classification output.
- Obtained **43% accuracy with ViT** and **97% accuracy with SWIN Transformer** for same image dataset with on 50 epochs.
- Inferenced **Flan T5**, **Vicuna 7B** and **Llama 3.2 3B** with high **Rouge-N of 0.73** with Llama. Integrated it as Language Model in BLIP2

Automation Engineer

Sep 2019 - Aug 2022

Bosch Global Software Technologies Pvt. Ltd.

Leadership Exposure:

- Led a team of three engineers to develop tools enabling analysis of test results and report generation **reducing time by 40%**.
- Spearheaded the development of an Azure CAN Dashboard, that enables data transfer from ECUs to **Azure Cloud for CAR VR**.

RF Emission Analysis and Harmonic Anomaly Detection for Automotive Conformance

- Deployed a centralized **React web app** to view and store RF Emission data for conformance analysis using **Django**.
- Attained **85% accuracy** in anomaly detection for RF Harmonics using **BiLSTM Autoencoder**.
- Saved 90% of time** by generating plots and reports automatically.

Smart Inventory Tracking Using Asset Detection and OCR

- Led implementation on **Raspberry Pi 3B+** using **YoLov4-Tiny** and **Tesseract** for **asset detection** and labeling **saving 25 hours/week**
- Developed Django app, automating asset labeling and notifications, improved asset accountability, **reducing misplacement by 80%**.

Artificial Intelligence Intern

Jan 2019 - Mar 2019

ConnectM Technology Solutions Pvt. Ltd.

Truck Driver Assistance System with Safety Constraints using Facial Recognition and Voice Assistance ChatBot.

- Built voice recognition and face recognition systems using **CMU Pocket Sphinx** and **Haar Cascade** and **AWS Face Rekognition**.
- Trained **drowsiness detection** using Facial **Landmark detection** and made sure modules can be deployed on **Edge devices**.
- Seamlessly integrated the modules onto a **Raspberry Pi 3B**, ensuring optimal performance, user experience, and road safety.

EDUCATION

Masters in **Artificial Intelligence (Robotics and Autonomous Systems)**, *Arizona State University*

3.97 GPA

Bachelor of Engineering in **Electronics and Communication Engineering**, *PES Institute of Technology*

3.20 GPA

TECHNICAL SKILLS

Programming & Scripting: Python, C & Embedded C, MATLAB/Simulink, C++, Java, JavaScript, HTML, Bash

Interests: Computer Vision, Deep Learning, Natural Language Processing, Generative AI, Large Language Models, Transformers

Tools and Platforms: Pandas, NumPy, TensorFlow, Keras, PyTorch, OpenCV2, Scikit-learn, TensorRT, Onnx, TFLite, SQL, GIT, Docker

Certifications:

- Generative AI with Large Language Models, AWS and DeepLearning.AI, Sep 2024 | Skills: SageMaker, LLMs, PEFT, Quantization
- Finetuning Large Language Models, DeepLearning.AI, Jul 2024 | Skills: Large Language Models (LLMs), Transformers, PEFT, LoRA
- LangChain for LLM Application Development, DeepLearning.AI, Dec, 2024 | Skills: LangChain, Vector Databases, Prompt Engineer
- Introducing Multimodal Llama 3.2, DeepLearning.AI, Dec 2024 | Skills: Llama 3.2, Ollama, Meta AI, Multimodal models, Llama agent
- Functions, Tools and Agents with LangChain!, DeepLearning.AI, Dec, 2024 | Skills: LangChain, Tools, Pydantic classes, Agents
- Building and Evaluating Advanced RAG, DeepLearning.AI, Jul 2024 | Skills: Large Language Models (LLM), RAG
- How Diffusion Models Work! DeepLearning.AI, Jul 2024 | Skills: Diffusion Models, Stable Diffusion, Image Generation
- Machine Learning in Production, DeepLearning.AI, Jul 2024 | Skills: MLOps
- DeepLearning.AI TensorFlow Developer, DeepLearning.AI, Jul 2024 | Skills: TensorFlow, Keras, Algorithms, OpenCV
- MLOps | Machine Learning Operations, Duke University, Jul 2023 | Skills: MLflow, DevOps, AWS SageMaker, MLOps
- Deep Learning Specialization, DeepLearning.AI, May 2020 | Skills: Image Processing/Recognition, Data Analysis, Neural Networks
- Machine Learning, DeepLearning.AI, Nov 2019 | Skills: Applied Mathematics, Machine Learning, Time Series, Pattern Recognition

PROJECTS

Negative CLIPort: Robotic Arm Control enhanced with Negative Language Goals | **Perception in Robots, ASU**

Spring 2023

- Improved **CLIPort** to handle a wider range of requirements by making it more robust to **negative language goals**.
- Training data augmented with goals that specify actions arm avoids, and loss function modified to penalize conflicting actions.

Medical Image Analysis for Lung Disease Classification and Detection | **Medical Image Processing, ASU**

Fall 2023

- Leveraged models/techniques like **Swin Transformer**, **Faster R-CNN**, **UPerNet**, and **U-Net** for classification, localization, and segmentation of Chest X Rays like **VinBigData CXR**, **SIIM-ACR Pneumothorax**, **MedMNISTv2** and **MINIJSRT Datasets**
- Developed a **multi-task model** to accurately analyze X-rays, achieving high performance in **metrics like AUC, IoU, FROC**.

Real-Time Hand Gesture Recognition System for Conference Calling and Media Control | **Embedded ML, ASU**

Fall 2023

- Utilized **Google's MediaPipe** framework to extract hand features from the image. Based on the app we developed 3 classifiers.
- Deployed ASL, conference call and media control apps on Raspberry Pi 4B. Achieved a high frame rate and 93% accuracy.

Facial Image Synthesis from Segmentation Maps/Sketches from GANs for Forensics | **Statistical Machine Learning, ASU**

Fall 2023

- Utilized CelebAMask-HQ dataset with **Pixel2Style2Pixel GAN** to transform sketches and segmentation maps to real images.
- The real face images generated with good accuracy and clarity and is useful for Detective work.

GSelect Branch Predictor and LRU-IPV cache replacement policy | **Advanced Computer Architecture, ASU**

Fall 2023

- Oversaw and worked with a group of three individuals while managing tasks. Researched and developed the state-of-the-art GSelect Branch predictor and LRU-IPV cache replacement strategy in C++ and Google Cloud Platform after understanding Gem5.